Worksheet 6: Week 5–9 May

Question 1.

$f(x) = \begin{cases} 4-x^2 & \text{as } x \le 1 \\ x-1 & \text{as } x > 1 \end{cases}$

- $f(1) = 3$
- $\lim_{x\to 1^-} 4-x^2 = 3 \neq \lim_{x \to 1^+} x-1 = 0$

$\therefore \lim_{x\to 1} f(x)$ doesn't exist

° f is discontinuous in $x=1$

Hence not continuous on $\mathbb{R}$.

Question 2.

1) $f(x) = 3^{x^2} e^{\sqrt{x}}$

$\Rightarrow f'(x) = (3^{x^2}\ln 3)(2x)e^{\sqrt{x}} + 3^{x^2} e^{\sqrt{x}} \times \frac{1}{2\sqrt{x}}$

2) $f(\theta) = \dfrac{\tan 2\theta}{1+\sec(\theta^2)}$

$\Rightarrow f'(\theta) = \dfrac{(2\sec^2 2\theta)(1+\sec(\theta^2)) - (\tan 2\theta)(2\theta)(\sec(\theta^2))(\tan\theta^2)}{[1+\sec(\theta^2)]^2}$

3) $f(y) = \sqrt[3]{1-\ln(1+2y)}$

$\Rightarrow f'(y) = \frac{1}{3}(1-\ln(1+2y))^{-\frac{2}{3}}\left(0 - \frac{2}{1+2y}\right)$

4) $f(x) = \tan^{-1}(e^{2x})$

$\Rightarrow f'(x) = \dfrac{1}{1+(e^{2x})^2} \times 2e^{2x}$

5) $f(x) = \sin^{-1}(1-4t)$

$\Rightarrow f'(x) = \dfrac{1}{\sqrt{1-(1-4t)^2}} \times -4$

6) $f(t) = 2\cos(\ln t) - 4\ln(1+\cos 6t)$

$\Rightarrow f'(t) = -2\sin(\ln t) \times \frac{1}{t} - (4)\dfrac{(-6\sin 6t)}{1+\cos 6t}$

15* $\frac{d}{du}(u)^{\sin^{-1}u}$ (Let $y = (u)^{\sin^{-1}u}$)

$\ln y = \sin^{-1}u \ln u$

$\frac{dy}{dx} = y\left[\arcsin u \ln u + \sin^{-1}u \times \frac{1}{u}\right]$

7) $x^3 \sin y + xy^4 = y + e$

$\Rightarrow 3x^2 \sin y + x^3\cos y \frac{dy}{dx} + y^4 + (x)(4y^3)(y') = \frac{dy}{dx} + 0$

$\Rightarrow (-1+x^3\cos y - 4xy^3)\frac{dy}{dx} = -y^4 - 3x^2\sin y$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{-y^4 - 3x^2 \sin y}{x^3\cos y - 4xy^3 - 1}$

8) $y = 2\sin(x^3) + \sin^3(2x)$

$\Rightarrow y' = 2\cos(x^3) \times 3x^2 + 3(\sin(2x))^2 \times \cos 2x \times 2$

9) $f(a) = 2^{3a} + (3a)^2 + 3^{2a}$

$\Rightarrow f'(a) = 2^{3a}\ln 2 \times 3 + (2)(3a)(3) + 3^{2a}\ln 3 \times 2$

10) $f(z) = z^{\cos z}$

Let $y = z^{\cos z}$

then $\ln y = \cos z \ln z$

$\Rightarrow \frac{1}{y}\frac{dy}{dx} = (-\sin z)\ln z + \cos z \times \frac{1}{z}$

$\Rightarrow \frac{dy}{dx} = y\left[-\sin z \ln z + \frac{\cos z}{z}\right]$

11) $f(x) = \sinh(\sin x)$

$\Rightarrow f'(x) = \cosh(\sin x) \times \cos x$

12) $f(x) = \cosh\left(\frac{1}{x}\right)$

$\Rightarrow f'(x) = -\sinh\left(\frac{1}{x}\right) \times -x^{-2}$

13) $f(x) = \ln\left(\sin(2x^2)\sqrt{x^2-1}\right)$

$\Rightarrow f'(x) = \dfrac{4x\cos(2x^2)(\sqrt{x^2-1}) + \sin(2x^2) \times \frac{2x}{2\sqrt{x^2-1}}}{\sin(2x^2)\sqrt{x^2-1}}$

14) $F(t) = \dfrac{t^2}{\sqrt{t^2-1}}$

$\Rightarrow F'(t) = \dfrac{(2t)(\sqrt{t^2-1}) - (t^2)\frac{2t}{2\sqrt{t^2-1}}}{(\sqrt{t^2-1})^2}$

15) $g(u) = \sin(u)^{\sin^{-1}u}$

$\Rightarrow g'(u) = \cos(u)^{\sin^{-1}u} \cdot \frac{d}{du}(u)^{\sin^{-1}u}$ *

So $g'(u) = (\cos u)^{\sin^{-1}u} \times u^{\sin^{-1}u}\left[\arcsin u \ln u + \sin^{-1}u \times \frac{1}{u}\right]$

Question 3

(1) $f(x) = \ln(4+x^2)$ $\quad 4+x^2>0$

- $D_f = \mathbb{R}$

$$f'(x) = \frac{2x}{4+x^2} = 0 \Rightarrow x = 0 \in D_f$$

- Critical number: $c = 0$

(2) $f(x) = \ln(x^2+1)$ $\quad x \in [0,1]$

f is continuous on $[0,1]$

$$f'(x) = \frac{2x}{x^2+1}$$, f is diff op $(0,1)$

$\exists$ n $c \in (0,1)$ so dat

$$f'(c) = \frac{f(1)-f(0)}{1-0}$$

$$\Rightarrow \frac{2c}{c^2+1} = \frac{\ln 2}{1}$$

$$\Rightarrow 2c = (\ln 2)(c^2+1)$$

$$\Rightarrow 0 = (\ln 2)c^2 + \ln 2 - 2c$$

$$\Rightarrow 0 = (\ln 2)c^2 - 2c + \ln 2$$

$$\Rightarrow c = \frac{-(-2) \pm \sqrt{4-(4)(\ln 2)^2}}{2\ln 2}$$

$$= \frac{2 \pm 2\sqrt{1-(\ln 2)^2}}{2\ln 2}$$

$$= \frac{1 \pm \sqrt{1-(\ln 2)^2}}{\ln 2}$$

But $c \in (0,1)$

- So $c = \dfrac{1-\sqrt{1-(\ln 2)^2}}{\ln 2}$

(use calculator) $= 0,4028061136$

(3) $f(x) = x^3 e^{-2x}$

$D_f = \mathbb{R}$

$$f'(x) = 3x^2 e^{-2x} + x^3 e^{-2x} \times -2$$

$$= \underset{>0}{e^{-2x}}\ \underset{>0}{x^2}\ \underset{<0}{(3-2x)} < 0$$

$$\Rightarrow 3-2x<0 \Rightarrow x > \tfrac{3}{2}$$

- f is decreasing on $(3/2, \infty)$

(4) $f(x) = 4x^4 - \ln x$

$D_f = (0,\infty)$

$$f'(x) = 16x^3 - \frac{1}{x}$$

$$= \frac{16x^4-1}{x}$$

$$= \frac{(4x^2-1)(4x^2+1)}{x}$$

$$= \frac{(2x-1)(2x+1)\underset{>0}{(4x^2+1)}}{\underset{>0}{x}} \quad (D_f = (0,\infty))$$

$\therefore f'(x) > 0$ $\quad y = 4x^2-1$

$\Rightarrow (2x-1)(2x+1) > 0$

$\Rightarrow x \in (-\infty, -\tfrac{1}{2}) \cup (\tfrac{1}{2}, \infty)$ $\quad -\tfrac{1}{2} \quad \tfrac{1}{2}$

- f is increasing on $(\tfrac{1}{2}, \infty)$ $(D_f = (0,\infty))$

(5) $f(x) = x e^{-2x^2}$

$$f'(x) = e^{-2x^2} + x e^{-2x^2} \times -4x$$

$$= \underset{>0}{e^{-2x^2}}(1-4x^2)$$

$$\Rightarrow f'(x) = 0 \Rightarrow 1-4x^2 = 0$$

$$\Rightarrow (1-2x)(1+2x) = 0$$

f': neg $\;-\tfrac{1}{2}\;$ pos $\;\tfrac{1}{2}\;$ neg $\quad (D_f = \mathbb{R})$

- f has a local min at $x = -\tfrac{1}{2}$

namely $f(-\tfrac{1}{2}) = -\tfrac{1}{2} e^{-2(+\frac{1}{4})}$

$$= -\frac{1}{2\sqrt{e}}$$

- f has a local max at $x = \tfrac{1}{2}$

namely

$$f(\tfrac{1}{2}) = \tfrac{1}{2} e^{-2(\frac{1}{4})} = \tfrac{1}{2} e^{-\frac{1}{2}}$$

$$= \frac{1}{2\sqrt{e}}$$

(6) $f(x) = e^x \sin x$, $x \in [-2\pi, 2\pi]$

$D_f = \mathbb{R}$

$$f'(x) = e^x \sin x + e^x \cos x$$

$$= e^x(\sin x + \cos x)$$

$$f''(x) = e^x(\sin x + \cos x) + e^x(\cos x - \sin x)$$

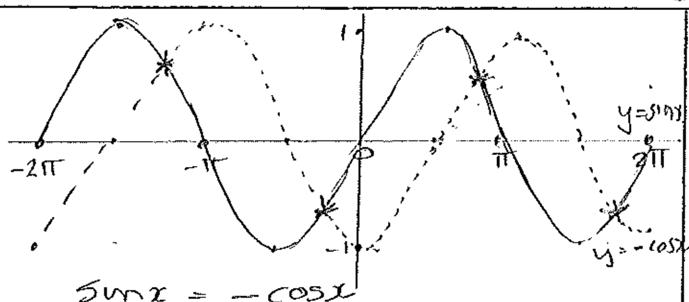
$$= e^x \sin x + e^x \cos x + e^x \cos x - e^x \sin x$$

$$= 2e^x \cos x$$

$$f'(x) = 0 \Rightarrow \underset{>0}{e^x}(\sin x + \cos x) = 0$$

$$\Rightarrow \sin x + \cos x = 0$$

$$\Rightarrow \sin x = -\cos x$$



$\sin x = -\cos x$

$* \Rightarrow x = \frac{3\pi}{4},\ \frac{3\pi}{4}+\pi = \frac{7\pi}{4},\ \frac{3\pi}{4}-2\pi = -\frac{5\pi}{4},$

$\frac{7\pi}{4} - 2\pi = -\frac{\pi}{4}$

$\Rightarrow x \in \{-\frac{5\pi}{4}, -\frac{\pi}{4}, \frac{3\pi}{4}, \frac{7\pi}{4}\}$

$f''(x) = 2\cos x\, \underbrace{e^{x}}_{>0}$

$\therefore f''(-\frac{5\pi}{4}) = \cos(-\frac{5\pi}{4}) < 0$

$f''(-\frac{\pi}{4}) = \cos(-\frac{\pi}{4}) > 0$

$f''(\frac{3\pi}{4}) = \cos(\frac{3\pi}{4}) < 0$

$f''(\frac{7\pi}{4}) = \cos(\frac{7\pi}{4}) > 0$

- f has a local min at $x = -\frac{\pi}{4}$, $x = \frac{7\pi}{4}$
- f has a local max at $x = -\frac{5\pi}{4}$, $x = \frac{3\pi}{4}$

(7) $f(x) = e^{x^2-4x}$ on $[0,3]$

$D_f = \mathbb{R}$. f is continuous on $\mathbb{R}$. Therefore f is continuous on any closed interval.

$f'(x) = (2x-4)\underbrace{e^{x^2-4x}}_{>0}$

$f'(x) = 0 \Rightarrow 2x-4 > 0 \Rightarrow x = 2$

x	0	2	3
$f(x)$	1	0,018	0,049

- The absolute minimum is $f(2) \approx 0{,}018$
- The absolute maximum is $f(0) = 1$

(8) $f(x) = x + \ln(x^2+1)$

$f'(x) = 1 + \frac{2x}{x^2+1}$

$f''(x) = \frac{2(x^2+1) - (2x)(2x)}{(x^2+1)^2}$

$= \frac{2-2x^2}{(x^2+1)^2}$

$f''(x) < 0$

$\Rightarrow 2(1-x^2) < 0$ as $(x^2+1)^2 > 0$

$\Rightarrow x \in (-\infty,-1) \cup (1,\infty)$ $\quad y = 1-x^2$

- f is concave downward on $(-\infty,-1)$ and on $(1,\infty)$

(9) $f(x) = \ln(1-x^2)$

(i) $1-x^2 > 0$

$\Rightarrow x \in (-1,1)$

- $D_f = (-1,1)$

(ii) • Horizontal asymptotes (None)

Since: $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$ does not exist.

- Vertical asymptotes

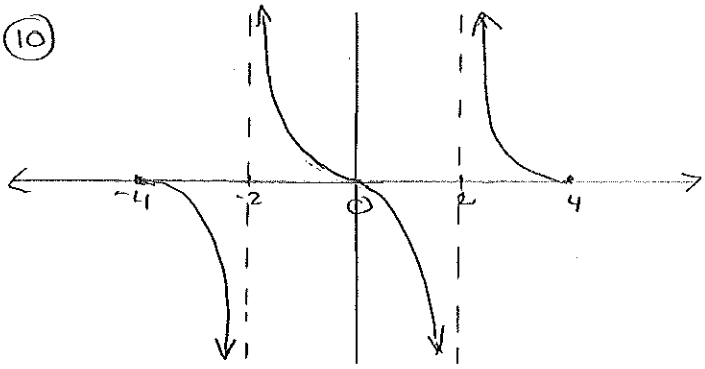
$x=-1$ is a vertical asymptote since

$\lim_{x\to-1^+} f(x) = \lim_{x\to-1^+} \ln(1-x^2) = -\infty$

$x=1$ is a vertical asymptote since

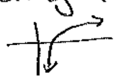
$\lim_{x\to1^-} f(x) = \lim_{x\to1^-} \ln(1-x^2) = -\infty$

(10)



Question 4

1. $\lim_{x\to 0^+} \ln x$ (using a graph)



$= -\infty$

2. $\lim_{x\to 2^+} \frac{x}{2-x} = -\infty$

(because if $x \to 2^+ \Rightarrow x > 2$
$\Rightarrow 2 - x < 0$)

3. $\lim_{x\to\infty} \frac{e^{4x}}{\ln x}$ $\left(\frac{\infty}{\infty}\right)$

$= \lim_{x\to\infty} \frac{4e^{4x}}{\frac{1}{x}} = \lim_{x\to\infty} 4xe^{4x} = +\infty$

4. $\lim_{x\to\infty} \arctan x$ ($\arctan x$
$= \operatorname{bgtan} x$
$= \tan^{-1} x$)

$= \pi/2$

(using graph)

5. $\lim_{x\to -\infty} \frac{\sqrt{4x^2-x}}{x}$

$= \lim_{x\to-\infty} \frac{|x|\sqrt{4-\frac{1}{x}}}{x}$ $\left(\sqrt{x^2} = |x| = -x \ (x<0)\right)$

$= \lim_{x\to-\infty} \frac{-x\sqrt{4-\frac{1}{x}}}{x} = -\sqrt{4} = -2$

OR

$\lim_{x\to-\infty} \frac{\sqrt{4x^2-x}}{x}$ $\left(\frac{\infty}{-\infty}\right)$

$= \lim_{x\to-\infty} \frac{\frac{8x-1}{2\sqrt{4x^2-x}}}{1} = \lim_{x\to-\infty} \frac{8x-1}{2|x|\sqrt{4-\frac{1}{x}}}$

$= \lim_{x\to-\infty} \frac{x(8-\frac{1}{x})}{2(-x)\sqrt{4-\frac{1}{x}}} = \lim_{x\to-\infty} \frac{8-\frac{1}{x}}{-2\sqrt{4-\frac{1}{x}}}$

$= \frac{8}{-2\sqrt{4}} = -2$

6. $\lim_{x\to-\infty} \frac{e^{4x}}{\ln|x|} = 0$

(because $\lim_{x\to-\infty} e^{4x} = 0$

and $\lim_{x\to-\infty} \ln|x| = -\infty \neq 0$

7. $\lim_{x\to\infty} \frac{x}{x-1} = \lim_{x\to\infty} \frac{1}{1-1/x} = 1$

8. $\lim_{x\to\infty} \frac{x^2-1}{x^3-4x} = \lim_{x\to\infty} \frac{\frac{x^2}{x^3} - \frac{1}{x^3}}{1 - \frac{4x}{x^3}}$

$= \lim_{x\to\infty} \frac{\frac{1}{x} - \frac{1}{x^3}}{1 - \frac{4}{x^2}} = \frac{0}{1} = 0$

9. $\lim_{x\to 0} \frac{e^{2x}-1}{x}$ $\left(\frac{0}{0}\right)$

$= \lim_{x\to 0} \frac{2e^{2x}}{1} = 2e^0 = 2$

10. $\lim_{x\to 0^+} \left(1+\frac{2}{x}\right)^x$ (∞^0)

Let $y = \left(1+\frac{2}{x}\right)^x$ then $\ln y = x\ln\left(1+\frac{2}{x}\right)$

$\lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} x\ln\left(1+\frac{2}{x}\right)$ $(0\times\infty)$

$= \lim_{x\to 0^+} \frac{\ln\left(1+\frac{2}{x}\right)}{\frac{1}{x}}$ $\left(\frac{\infty}{\infty}\right)$

$= \lim_{x\to 0^+} \frac{-\frac{2}{x^2}}{1+\frac{2}{x}} \div -\frac{1}{x^2}$

$= \lim_{x\to 0^+} \frac{-\frac{2}{x^2}}{1+\frac{2}{x}} \times -\frac{x^2}{1}$

$= \lim_{x\to 0^+} \frac{2}{1+\frac{2}{x}} = 0$

$\therefore \lim_{x\to 0^+} y = e^0 = 1$

11. $\lim_{x\to\infty} \left(1-\frac{2}{x}\right)^{3x}$ (1^∞)

Let $y = \left(1-\frac{2}{x}\right)^{3x}$ then $\ln y = 3x\ln\left(1-\frac{2}{x}\right)$

$\lim_{x\to\infty} \ln y = \lim_{x\to\infty} 3x\ln\left(1-\frac{2}{x}\right)$ $(\infty\times 0)$

$= \lim_{x\to\infty} \frac{3\ln\left(1-\frac{2}{x}\right)}{\frac{1}{x}}$ $\left(\frac{0}{0}\right)$

$= \lim_{x\to\infty} \frac{3\times +\frac{2}{x^2}}{\left(1-\frac{2}{x}\right)} \div -\frac{1}{x^2}$

$= \lim_{x\to\infty} \frac{-(3)(2)}{1-\frac{2}{x}} = -6$

$\therefore \lim_{x\to\infty} y = e^{-6}$

12. $\lim_{x\to\infty} \frac{x^2+1}{\ln(x^2+1)}$ $\left(\frac{\infty}{\infty}\right)$

$= \lim_{x\to\infty} \frac{2x}{\frac{2x}{x^2+1}} = \lim_{x\to\infty} \frac{1}{x^2+1}$

$= 0$